

# Experiment-06: Vacuum Cleaner Problem using Python

## Aim

To develop a Python program that simulates the **Vacuum Cleaner Problem** using a **Simple Reflex Agent**, where the vacuum cleans dirty rooms and moves until all rooms are clean.

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## Problem Statement

A vacuum cleaner operates in two rooms:

- **Room A**
- **Room B**

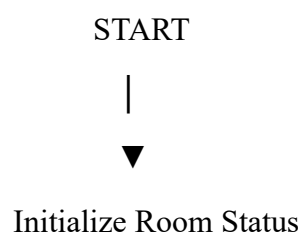
## Rules

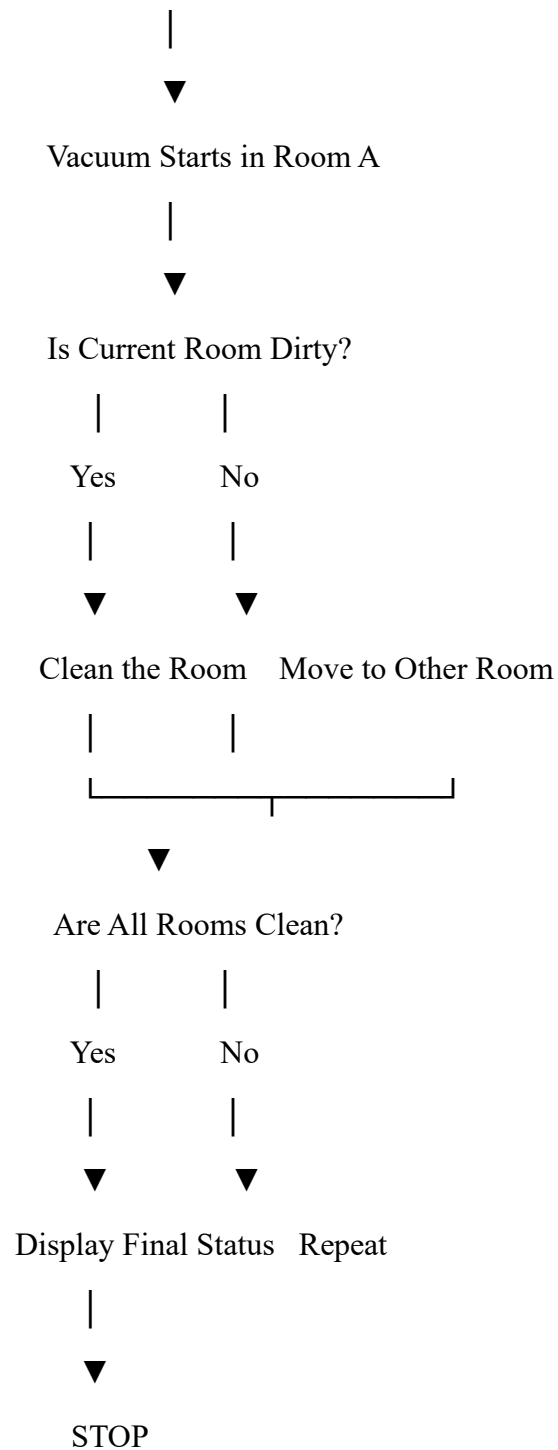
- If the current room is **Dirty**, the vacuum cleans it.
  - If the current room is **Clean**, it moves to the other room.
  - The process continues until **both rooms are clean**.
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## Algorithm

1. Start with the vacuum in Room A.
  2. Check the status of the current room.
  3. If the room is dirty, clean it.
  4. If the room is clean, move to the other room.
  5. Repeat Steps 2–4 until both rooms are clean.
  6. Display the actions performed and the final room status.
  7. Stop.
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## Flowchart






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## Python Code

# Vacuum Cleaner Problem using Simple Reflex Agent

```
def vacuum_cleaner(room_status, location):
```

```
print("Initial Room Status:")
```

```
print(room_status)
```

```
print()
```

```
while True:
```

```
    if room_status["A"] == "Clean" and room_status["B"] == "Clean":
```

```
        print("\nAll rooms are clean.")
```

```
        break
```

```
    print("Vacuum is in Room", location)
```

```
    if room_status[location] == "Dirty":
```

```
        print("Room", location, "is Dirty.")
```

```
        print("Action: Cleaning Room", location)
```

```
        room_status[location] = "Clean"
```

```
    else:
```

```
        print("Room", location, "is already Clean.")
```

```
    if location == "A":
```

```
        location = "B"
```

```
    else:
```

```
        location = "A"
```

```
    print("Current Status:", room_status)
```

```
    print()
```

```
print("Final Room Status:")
```

```
print(room_status)

# Initial room conditions

rooms = {

    "A": "Dirty",

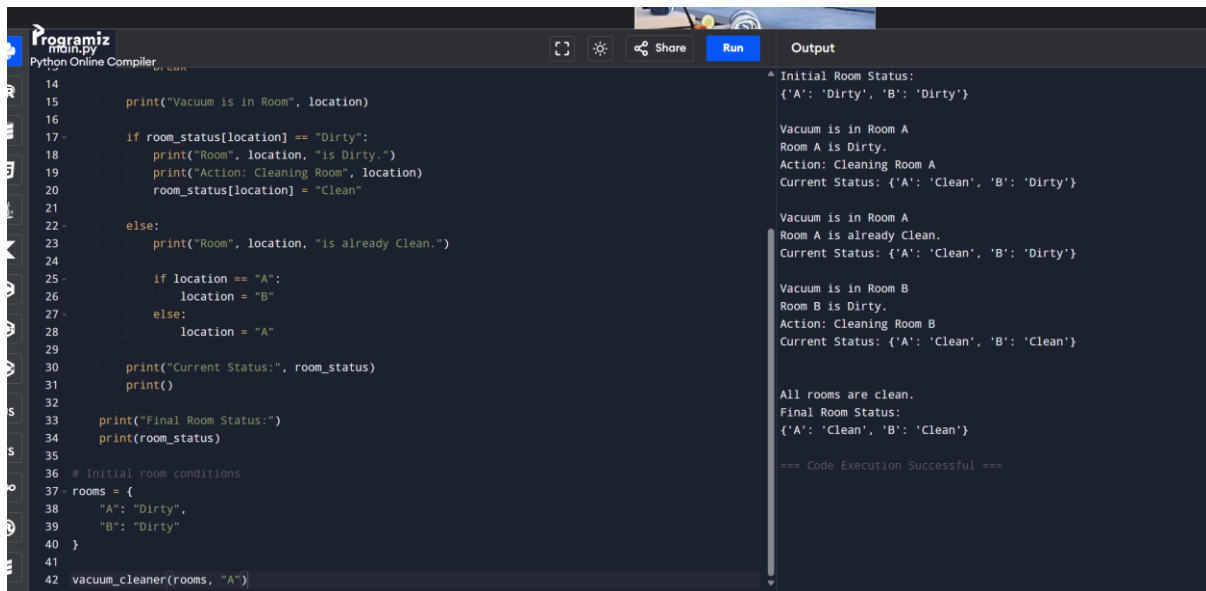
    "B": "Dirty"

}

vacuum_cleaner(rooms, "A")
```

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## Output



```
Programiz
Python Online Compiler
14 print("Vacuum is in Room", location)
15
16
17 if room_status[location] == "Dirty":
18     print("Room", location, "is Dirty.")
19     print("Action: Cleaning Room", location)
20     room_status[location] = "Clean"
21
22 else:
23     print("Room", location, "is already Clean.")
24
25     if location == "A":
26         location = "B"
27     else:
28         location = "A"
29
30 print("Current Status:", room_status)
31 print()
32
33 print("Final Room Status:")
34 print(room_status)
35
36 # Initial room conditions
37 rooms = {
38     "A": "Dirty",
39     "B": "Dirty"
40 }
41
42 vacuum_cleaner(rooms, "A")
```

Initial Room Status:  
{'A': 'Dirty', 'B': 'Dirty'}  
  
Vacuum is in Room A  
Room A is Dirty.  
Action: Cleaning Room A  
Current Status: {'A': 'Clean', 'B': 'Dirty'}  
  
Vacuum is in Room A  
Room A is already Clean.  
Current Status: {'A': 'Clean', 'B': 'Dirty'}  
  
Vacuum is in Room B  
Room B is Dirty.  
Action: Cleaning Room B  
Current Status: {'A': 'Clean', 'B': 'Clean'}  
  
All rooms are clean.  
Final Room Status:  
{'A': 'Clean', 'B': 'Clean'}  
  
=== Code Execution Successful ===

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## Result

The **Vacuum Cleaner Problem** was successfully implemented using a **Simple Reflex Agent** in Python. The vacuum cleaner sensed the condition of each room, cleaned dirty rooms, moved between rooms when necessary, and stopped only after both rooms were clean. This demonstrates the working of a simple intelligent agent based on condition–action rules.